

Farad Wahab

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Education

Oregon State University, B.S. Computer Science

Expected June 2028

GPA 3.6 · Coursework: Data Structures, Algorithms, Operating Systems, Databases, Software Engineering

Experience

Software Engineering Intern, Willamette Data Co.

Jun – Sep 2025

- Built an internal React dashboard that replaced a weekly spreadsheet report, saving the operations team about 4 hours a week.
- Moved 3 nightly data jobs from cron scripts to a Python pipeline with retries and alerts; failed runs went from about one a week to none.

Undergraduate Teaching Assistant, Data Structures, Oregon State University

Sep 2024 – Jun 2025

- Ran weekly lab sections of about 30 students and held office hours.
- Wrote autograder tests for two of the course's programming assignments.

Projects

Chess engine (C++, Python)

2024

- Minimax search with alpha-beta pruning reaching 6 ply in under 2 s (1.2M positions/s).
- Made search 9× faster by ordering captures and checks first.

Banking app (Java, Spring Boot, PostgreSQL, React)

2025

- Led the backend on a 3-person team: transfer logic, schema and authentication.
- Found and fixed a race condition with row-level locking; 0 balance mismatches across 10,000 concurrent test transfers.

Housing affordability dashboard (Python, PostGIS, Plotly, React)

2025

- Joined 4 public data sources into 15 years of county-level data for 3,100+ US counties.
- Built the interactive county map and price-to-income history charts.

Video sharing platform (React, Node.js, PostgreSQL, S3, FFmpeg)

2025

- Streamed uploads up to 2 GB to S3 and transcoded each into 3 resolutions in a background worker.

Skills

Languages: Python, Java, TypeScript, C++, SQL

Frameworks: React, Next.js, Node.js, Spring Boot, FastAPI

Data & tools: PostgreSQL, PostGIS, Pandas, Plotly, Git, Docker, AWS (S3, EC2), Linux